MINI ASSESSMENT - LAB REPORT

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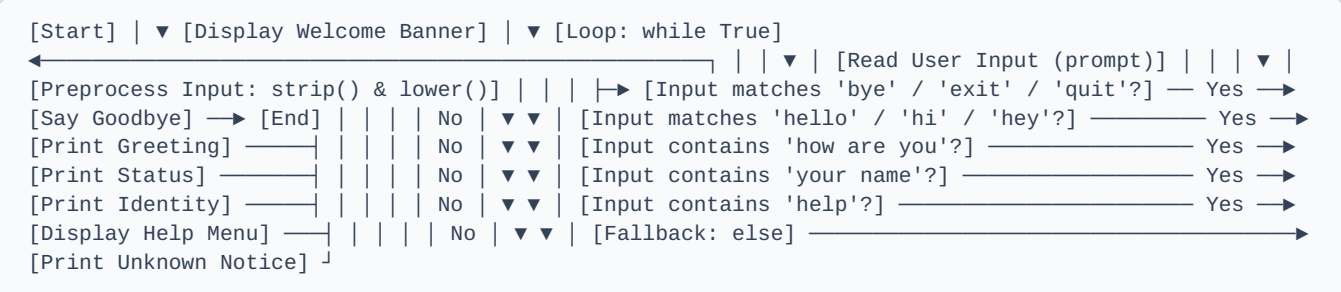
1. AIM

To design and develop an interactive, rule-based conversational chatbot in Python using conditional `if-elif-else` branch statements and an interactive execution loop to process user prompts and return contextually appropriate responses.

2. PROCEDURE

1. Initialize a continuous execution loop (`while True`) to sustain an ongoing conversational interface.
2. Prompt and read user input string via the standard `input()` method.
3. Standardize input format by trimming outer white space with `.strip()` and casting characters to lowercase via `.lower()` for uniform keyword evaluation.
4. Evaluate termination condition keywords (such as `'bye'`, `'exit'`, `'quit'`) to gracefully exit the execution loop.
5. Use sequential `if-elif` conditional branches to detect intent patterns (e.g., standard greetings, system inquiry, assistance queries).
6. Provide a reliable fallback mechanism in the final `else` block to manage unhandled or unrecognized inputs.

3. FLOWCHART



4. PYTHON CODE

```
def rule_based_bot():
    print("Bot: Hello! I am your assistant. Type 'bye' or 'exit' to quit.\n")

    while True:
        # Prompt and read user input
        user_input = input("You: ").strip().lower()

        # Check termination condition
        if user_input in ["bye", "exit", "quit"]:
            print("Bot: Goodbye! Have a great day!")
```

```
            break

        # Rule-based decision branches
        elif user_input in ["hello", "hi", "hey"]:
            print("Bot: Hello! How can I help you today?")

        elif "how are you" in user_input:
            print("Bot: I am doing great, thank you! How about you?")

        elif "your name" in user_input:
            print("Bot: I am a simple Python rule-based chatbot.")

        elif "help" in user_input:
            print("Bot: You can greet me, ask about my well-being, request my name, or type 'bye'
to exit.")

        else:
            print("Bot: I'm sorry, I don't understand that. Type 'help' to see what I can do.")

if __name__ == "__main__":
    rule_based_bot()
```

5. RESULT

```
Bot: Hello! I am your assistant. Type 'bye' or 'exit' to quit. You: hello Bot: Hello! How can I help
you today? You: how are you? Bot: I am doing great, thank you! How about you? You: what is your
name? Bot: I am a simple Python rule-based chatbot. You: can you dance? Bot: I'm sorry, I don't
understand that. Type 'help' to see what I can do. You: bye Bot: Goodbye! Have a great day!
```

6. CONCLUSION

> The rule-based conversational bot was successfully implemented using conditional branching structures (`if-elif-else`) inside an active `while` loop. The program demonstrates deterministic natural language pattern matching, handles conversation state termination, and gracefully falls back on unrecognized queries.